DO NOW — RECALL

Spiral model and Rapid Application Development

Name: _____ Class: _____ Date: _____

1. Name a key XP programming practice.

2. How does an Agile sprint help with changed requirements?

3. What is a prototype?

4. Why might risk analysis be valuable before developing an uncertain hardware-dependent system?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

For example pair programming, TDD or continuous integration.

2. Model answer

It lets the team adjust the next iteration using feedback.

3. Model answer

An early model used to explore and test an idea.

4. Model answer

It can reveal technical feasibility issues before expensive implementation.

Vocabulary — know these terms

Spiral model — Iterative methodology driven by risk identification and evaluation.

risk analysis — Evaluate probability/impact of potential problems.

RAD — Rapid Application Development using frequent prototypes and feedback.

prototype — Early working or mock representation for evaluation.

stakeholder — Person or group affected by or using the system.

NEW LEARNING

Read, annotate and explain

Spiral focuses on risk

Each cycle plans objectives, identifies or resolves major risks, develops/tests and reviews with stakeholders.

RAD is prototype-led

Users evaluate quickly produced models. Frequent prototyping accelerates understanding and can reveal unclear requirements.

Compare suitability

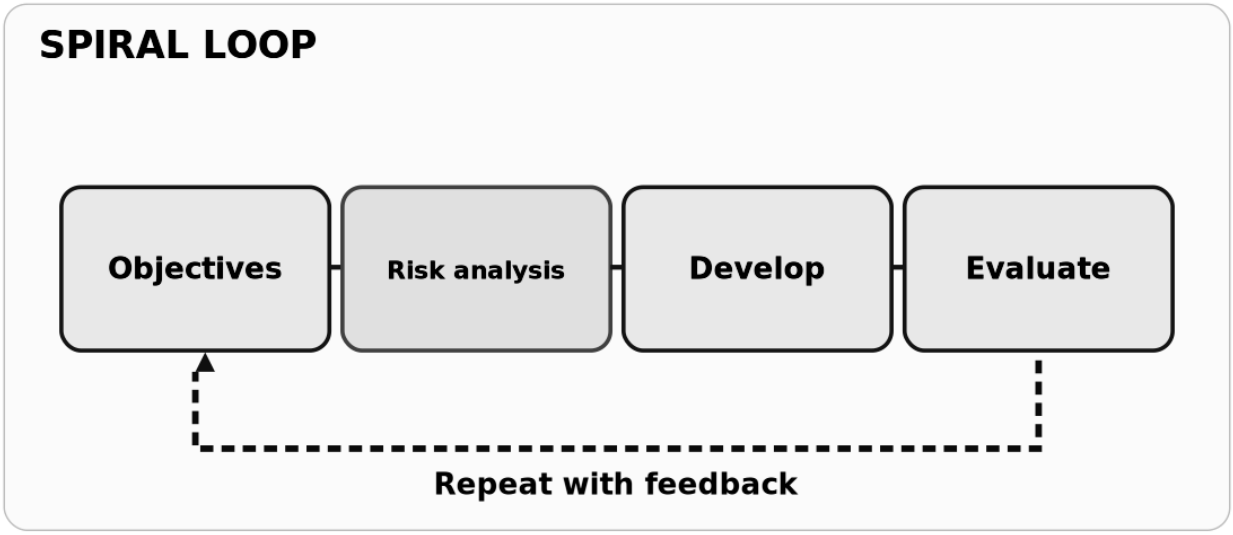
Spiral can suit large/high-risk systems but increases management cost; RAD relies on accessible users and skilled teams and may compromise architecture if rushed.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

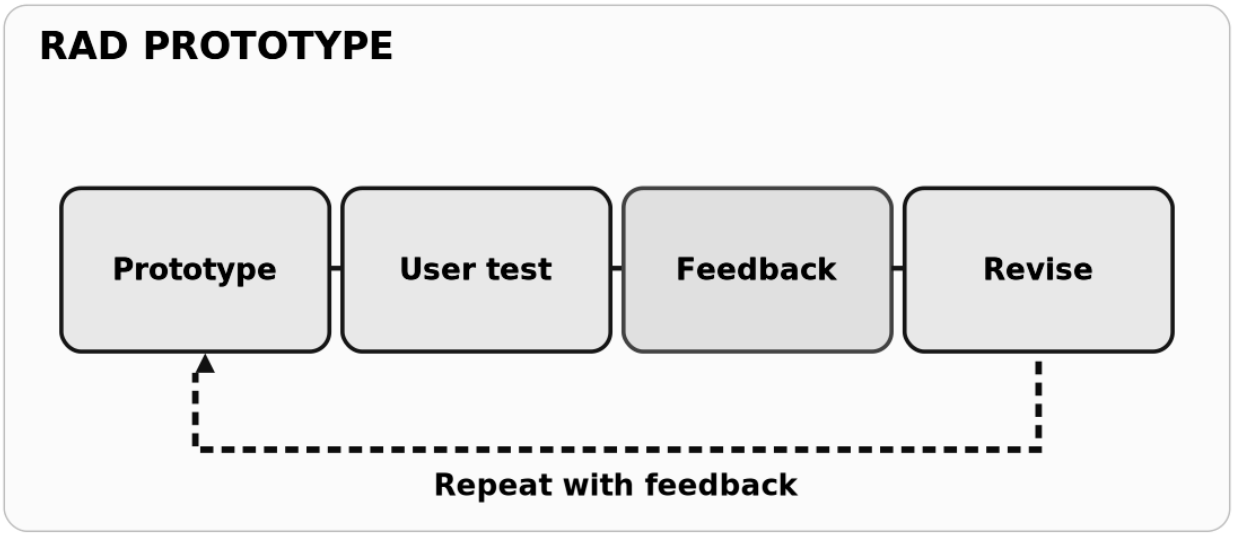
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



SPIRAL LOOP; Objectives to Risk analysis to Develop to Evaluate

New Learning

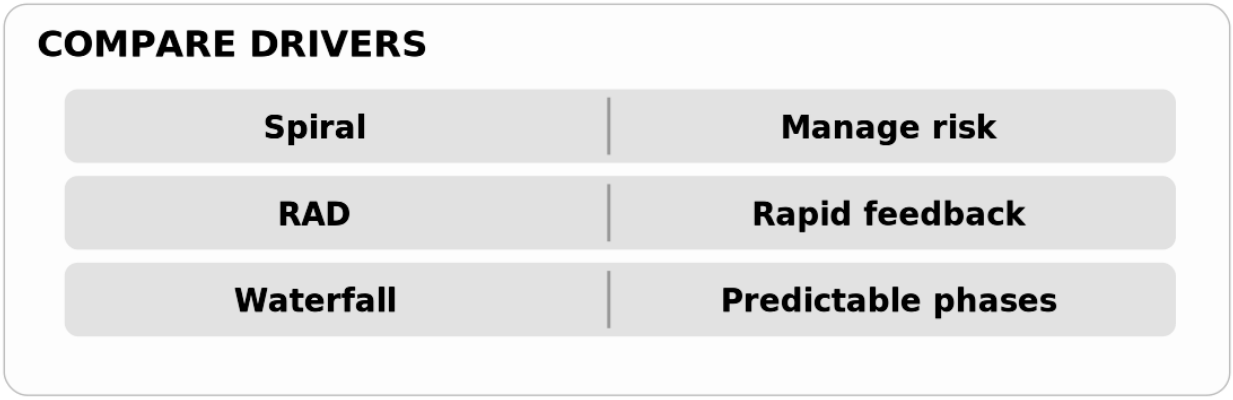


RAD PROTOTYPE; Prototype to User test to Feedback to Revise

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



COMPARE DRIVERS: Spiral Manage risk; RAD Rapid feedback; Waterfall Predictable phases

I DO — WORKED EXAMPLE

Follow the worked method

A satellite control project has major safety and technology uncertainties. Recommend a method.

Predict first

Spiral is appropriate because major technical and safety risks need assessment at each cycle.

Model and explain

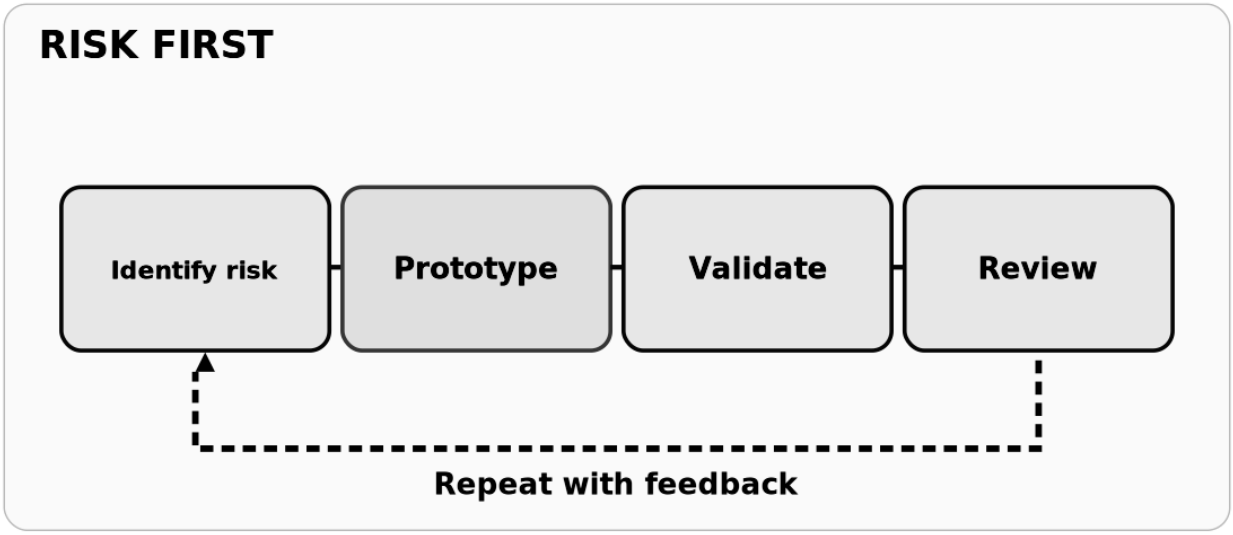
It demands experienced risk assessment and carries greater time/management overhead than simpler methods.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A satellite control project has major safety and technology uncertainties. Recommend a method.

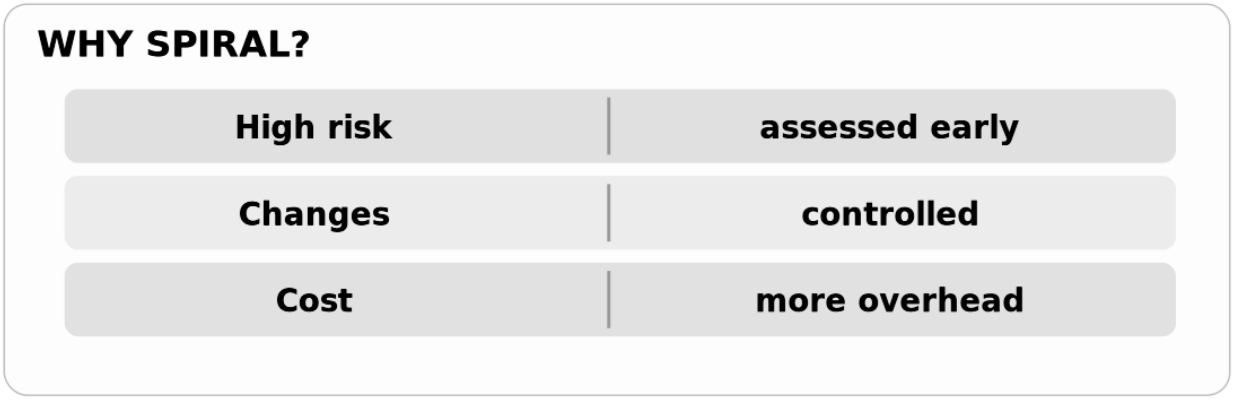
Spiral is appropriate because major technical and safety risks need assessment at each cycle.



RISK FIRST; Identify risk to Prototype to Validate to Review

A satellite control project has major safety and technology uncertainties. Recommend a method.

Spiral is appropriate because major technical and safety risks need assessment at each cycle.



WHY SPIRAL?: High risk assessed early; Changes controlled; Cost more overhead

WE DO — GUIDED PRACTICE

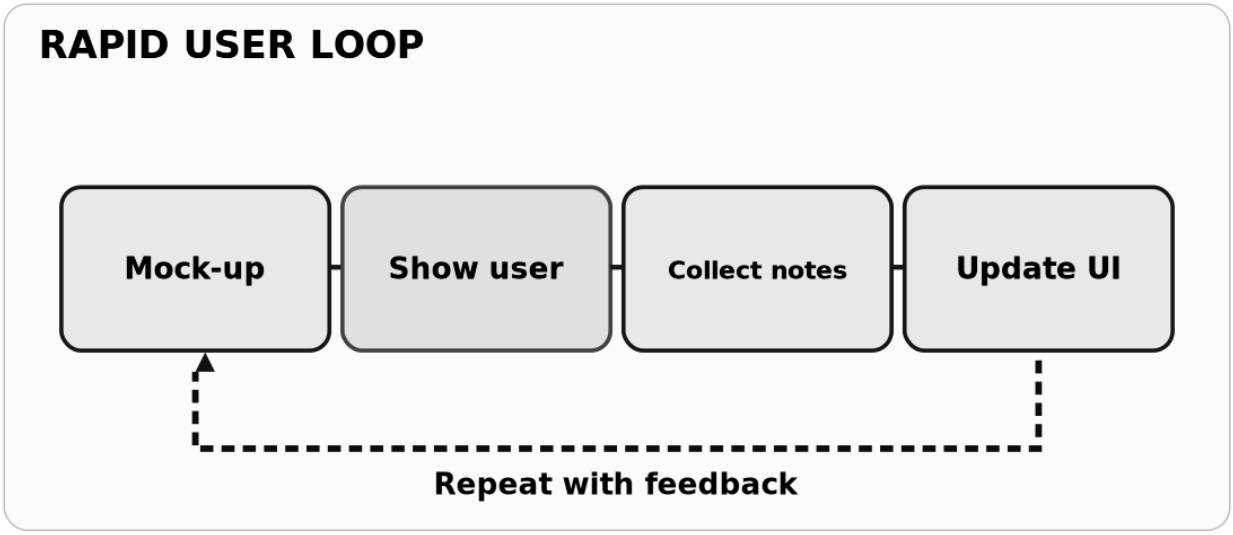
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A café needs a quick booking interface but cannot clearly describe its desired layout. Recommend a method.

RAD suits evolving interface requirements because the café can see and comment on working prototypes.



RAPID USER LOOP; Mock-up to Show user to Collect notes to Update UI

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first RAD suits evolving interface requirements because the café can see and comment on working prototypes.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State one key characteristic and one disadvantage of both Spiral and RAD.

Question 2

A transport company is building an experimental driverless control system. Compare Spiral with RAD and justify a choice.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A booking website must launch rapidly and users are available daily. Explain three benefits and two risks of RAD.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Spiral development and RAD are identical because both can repeat stages.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

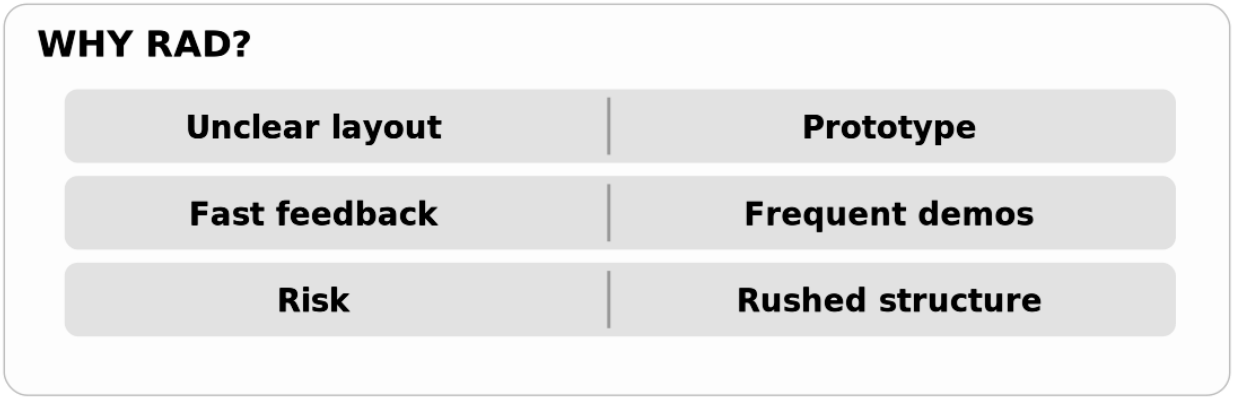
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A café needs a quick booking interface but cannot clearly describe its desired layout. Recommend a method.

RAD suits evolving interface requirements because the café can see and comment on working prototypes.



WHY RAD?: Unclear layout Prototype; Fast feedback Frequent demos; Risk Rushed structure

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model and explain

A risk is that repeated quick changes may weaken maintainability without time for robust design.

IMPROVE: Explain one correction to your own reasoning.